# EXPERIMENT NO: 19

Experiment to understand Logistic Regression for a given dataset.

## Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
print(df)
print(df.head())
features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
print(features)
print(label)
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
for i in range(1,401):

x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.
2,random_state=42)
  model=LogisticRegression()
  model.fit(x_train,y_train)
  train_score=model.score(x_train,y_train)
  test_score=model.score(x_test,y_test)
  if test_score>train_score:
      print("Test {} Train {} Random State
{}".format(test_score,train_score,i))
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.
2,random_state=42)
finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)
print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))
from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

## Output:

```
      User ID  Gender  Age  EstimatedSalary  Purchased
0    15624510    Male   19            19000          0
1    15810944    Male   35            20000          0
2    15668575  Female   26            43000          0
3    15603246  Female   27            57000          0
4    15804002    Male   19            76000          0
..        ...     ...  ...              ...        ...
395  15691863  Female   46            41000          1
396  15706071    Male   51            23000          1
397  15654296  Female   50            20000          1
398  15755018    Male   36            33000          0
399  15594041  Female   49            36000          1

[400 rows x 5 columns]
   User ID  Gender  Age  EstimatedSalary  Purchased
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1  15810944    Male   35            20000          0
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3  15603246  Female   27            57000          0
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0.7375
              precision    recall  f1-score   support

           0       0.83      0.90      0.86       250
           1       0.80      0.70      0.75       150

    accuracy                           0.82       400
   macro avg       0.82      0.80      0.81       400
weighted avg       0.82      0.82      0.82       400
```